2.4 Two Marks Questions & Answers

Question 26. What do you mean by boundary value problem? **Apr.'11**

Answer . In physical problems, we always seek a solution of the differential equation which satisfies some specified conditions known as the boundary conditions. The differential equation together with these boundary conditions, constitute a boundary value problem.

Question 27. Write down the laplace equation in two dimension. **Apr.'11**

Answer . Let $u(x, y)$ be the function of two independent variables x and y then the Laplace equation of u is given by $u_{xx} + u_{yy} = 0$.

Question 28. Explain the method of separation of variables. **Nov.'14**

Answer . In this method, collect all functions of x and dx one side and other side the functions of y and dy from the given equation then the variables are said to be separable. Thus the general form of such an equation is $f(x)dx = f(y)dy$. Integrating both sides we get $\int f(x)dx = \int f(y)dy + c$ as a solution.

Question 29. What do you mean by initial value problems?

Answer . By solving the ordinary differential equations, the solution consists of as many as arbitrary constants. To determine the value of such constants by putting initial values called as initial value problems.

Question 30. Classify the given equation $u_{xx} - 3u_{xy} + u_{yy} = 0$.

Solution . Here $A = 1, B = -3, C = 1$

$$\therefore \quad B^2 - 4AC = (-3)^2 - 4.1.1 = 9 - 4 = 5 > 0.$$

Hence the given equation is hyperbolic.

Question 31. Classify the given equation $u_{xx} - 2u_{xy} + u_{yy} = 0$.

Solution . Here $A = 1$, $B = -2$, $C = 1$

$$\therefore \quad B^2 - 4AC = (-2)^2 - 4.1.1 = 0.$$

Hence the given equation is parabolic.

Question 32. Classify the given partial differential equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2$.

Solution . Here $A = 1$, $B = 0$, $C = 1$.

$$\therefore \quad B^2 - 4AC = 0 - 4.1.1 = -4 < 0. \quad \text{It is elliptic.}$$

Question 33. Classify the given partial differential equation $4u_{xx} + 4u_{xy} + u_{yy} - 6u_x - 8u_y - 16u = 0$.

Solution . Here $A = 4$, $B = 4$, $C = 1$.

$$\therefore \quad B^2 - 4AC = (4)^2 - 4.4.1 = 0. \quad \text{It is parabolic.}$$

Question 34. Classify the given equation $f_{xx} + k^2 f_{yy} = 0$.

Solution . Here $A = 1$, $B = 0$, $C = k^2$

$$\therefore \quad B^2 - 4AC = 0 - 4.1.k^2 = -4k^2.$$

If $k = 0$ then $B^2 - 4AC = 0$, the given equation is parabolic.
If $k \neq 0$ then $B^2 - 4AC < 0$, the given equation is elliptic.

Question 35. Classify the given equation $u_{xx} + xu_{yy} = 0$. Nov.'14

Answer . Here $A = 1$, $B = 0$, $C = x$

$$\therefore \quad B^2 - 4AC = 0 - 4.1.x = -4x.$$

If $x = 0$ then $B^2 - 4AC = 0$, the given equation is parabolic.
If $x > 0$ then $B^2 - 4AC < 0$, the given equation is elliptic.
If $x < 0$ then $B^2 - 4AC > 0$, the given equation is hyperbolic.

Question 36. Classify the given equation $yu_{xx} + xu_{yy} = 0$. When $x > 0$, $y > 0$.

Solution . Here $A = y$, $B = 0$, $C = x$

$$\therefore \; B^2 - 4AC = 0 - 4yx = -4yx = \text{negative when } x > 0, y > 0$$

Hence the given equation is elliptic.

Question 37. Classify the given equation $x^2 f_{xx} + (1 - y^2) f_{yy} = 0$. When $-1 < y < 1$, $-\infty < x < \infty$.

Solution . Here $A = x^2$, $B = 0$, $C = 1 - y^2$

$$\therefore \; B^2 - 4AC = 0 - 4.x^2.(1 - y^2) = 4x^2(y^2 - 1).$$

If $x = 0$ then $B^2 - 4AC = 0$. Thus the given equation is parabolic. Also x^2 is always positive in $-\infty < x < \infty$ and $y^2 - 1$ is always negative when y lies in $-1 < y < 1$. $\therefore$ $B^2 - 4AC = -ve$. Hence the given equation is elliptic.

Nov.'14

Question 38. Classify the following PDE

(i) $\dfrac{\partial u}{\partial t} = \alpha^2 \dfrac{\partial^2 u}{\partial x^2}$ One dimensional heat equation

(ii) $\dfrac{\partial^2 y}{\partial t^2} = c^2 \dfrac{\partial^2 y}{\partial x^2}$ One dimensional wave equation

(iii) $\dfrac{\partial^2 y}{\partial x^2} + \dfrac{\partial^2 y}{\partial y^2} = 0$ Two dimensional heat equation

Solution . (i) Given one dimensional heat equation $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$.

That is, $\alpha^2 \frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial t} = 0.$

Here $A = \alpha^2$, $B = 0$, $C = 0$

$$\therefore \; B^2 - 4AC = 0 - 4\alpha^2.0 = 0.$$

Hence the one dimensional heat equation is ***parabolic***.

(ii)Given one dimensional wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$.

$$\text{That is,} \quad c^2 \frac{\partial^2 y}{\partial x^2} - \frac{\partial^2 y}{\partial t^2} = 0$$

Here $A = c^2$, $B = 0$, $C = -1$

$$\therefore \quad B^2 - 4AC = 0 - 4c^2.(-1) = 4c^2 > 0.$$

Hence the one dimensional heat equation is ***hyperbolic.***

(iii)Given two dimensional heat equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Here $A = 1$, $B = 0$, $C = 1$.

$$\therefore \quad B^2 - 4AC = 0 - 4.1.1 = -4 < 0.$$

Hence the one dimensional heat equation is ***elliptic.***

Question 39. Write any four assumptions made when deriving the wave equation. **Nov.'06, Nov.'11, Nov.'10, Nov.'15**

Answer . The following assumptions are made when deriving the wave equation.

1. The mass per unit length of the string is constant.
2. The string is perfectly elastic and so it does not offer resistance to bending.
3. Tension is very large compared to weight of the string.
4. Since the tension T is so large so that the gravitational force and the friction may be neglected.

Question 40. Write down the appropriate solution of the vibration of the string equation. How is it chosen? (or) Why the most appropriate solution of the one dimensional wave equation contains sines and cosines terms? **Apr.'11, Apr.'15**

Answer . The appropriate solution of the vibration of the string is

$$y(x,t) = (A\cos px + B\sin px)(C\cos apt + D\sin apt).$$

When we deal with the vibration of the string, it should be perfectly elastic. Then the transverse vibrations of the string at any point x it must be in **periodic** in 't'. Hence the above solution in which consists of periodic functions in 't'. So the appropriate solution is in the sine and cosine terms.

Question 41. Write the boundary conditions and initial conditions for solving the vibration of string equation, if the string is subjected to initial displacement $f(x)$ and initial velocity $g(x)$. **Apr.'11**

Answer . Let the wave equation is $\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$. ...(1)

The initial and boundary conditions of (1) is given by

(i) $y(0,t) = 0$, for all $t > 0$. (ii) $y(l,t) = 0$, for all $t > 0$

(iii) $\left[\frac{\partial y}{\partial t}\right]_{t=0} = g(x);\ 0 < x < l$ (iv) $y(x,0) = f(x)$

Question 42. Write down one dimensional wave equation and its various possible solutions. **Nov.'10, Apr.'15**

Answer . The one dimensional wave equation are $y_{tt} = a^2 y_{xx}$. The possible solutions of this wave equation are,

$$\begin{aligned}
y(x,t) &= (c_1 x + c_2)(c_3 t + c_4) \\
y(x,t) &= \left(c_5 e^{px} + c_6 e^{-px}\right)\left(c_7 e^{apt} + c_8 e^{-apt}\right) \\
y(x,t) &= \left(c_9 \cos px + c_{10} \sin px\right)\left(c_{11} \cos apt + c_{12} \sin apt\right).
\end{aligned}$$

Question 43. Write all possible solutions of the transverse vibrations of a string. **Nov.'11, Apr.'12, Nov.'15**

Answer .

$$\begin{aligned}
y(x,t) &= (c_1 x + c_2)(c_3 t + c_4) \\
y(x,t) &= \left(c_5 e^{px} + c_6 e^{-px}\right)\left(c_7 e^{apt} + c_8 e^{-apt}\right) \\
y(x,t) &= \left(c_9 \cos px + c_{10} \sin px\right)\left(c_{11} \cos apt + c_{12} \sin apt\right).
\end{aligned}$$

Question 44. In the wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$, what does c^2 stand for? **(or)** What is the constant a^2 in one dimensional wave equation $y_{tt} = a^2 y_{xx}$? **Nov.'11, Apr.'12, Apr.'15, Nov.'15**

Answer . c^2 (or) $a^2 = \frac{T}{m}$. Where T is the tension and m is the mass of the string.

Question 45. Write the most suitable solution of the one dimensional wave equation **Apr.'14**

Answer . The most suitable solution of the one dimensional wave equation are

$$y(x,t) = (c_9 \cos px + c_{10} \sin px)(c_{11} \cos apt + c_{12} \sin apt).$$

Question 46. A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially in a position given by $y(x,0) = V_0 \sin^3\left(\frac{\pi x}{l}\right)$. If it is released from rest in this position, write the boundary conditions. Apr.'12

Answer . The boundary conditions are

$$(i)\ y(0,t) = 0, \quad \text{for all } t > 0$$

$$(ii)\ y(l,t) = 0, \quad \text{for all } t > 0$$

$$(iii)\ \left[\frac{\partial y}{\partial t}\right]_{t=0} = 0;\ 0 < x < l$$

$$(iv)\ y(x,0) = V_0 \sin^3\left(\frac{\pi x}{l}\right).$$

Question 47. A string is stretched and fastened at two points $x = 0$ and $x = l$ at a distance 'l' apart. Motion is started by displacing the string into the form $y = k(lx - x^2)$ from which it is released at time $t = 0$. Write down the boundary conditions in mathematical form. Apr.'14

Answer . The boundary conditions are

$$(i)\ y(0,t) = 0, \quad \text{for all } t > 0$$

$$(ii)\ y(l,t) = 0, \quad \text{for all } t > 0$$

$$(iii)\ \left[\frac{\partial y}{\partial t}\right]_{t=0} = 0;\ 0 < x < l$$

$$(iv)\ y(x,0) = k(lx - x^2).$$